



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Assignment # 4

Course Title:	VLSI Design	Course Code:	CPE434	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Programme Name:	BS Computer Engineering		
Semester:		Batch:		Date:	14 th May 2026
Last Date:	19 th May 2026		Maximum Marks:	12	

Question # 1:

(PLO3-CLO3-C5)

Design a 4-bit Up/Down-Counter at transistor level with minimum number of Transistors.